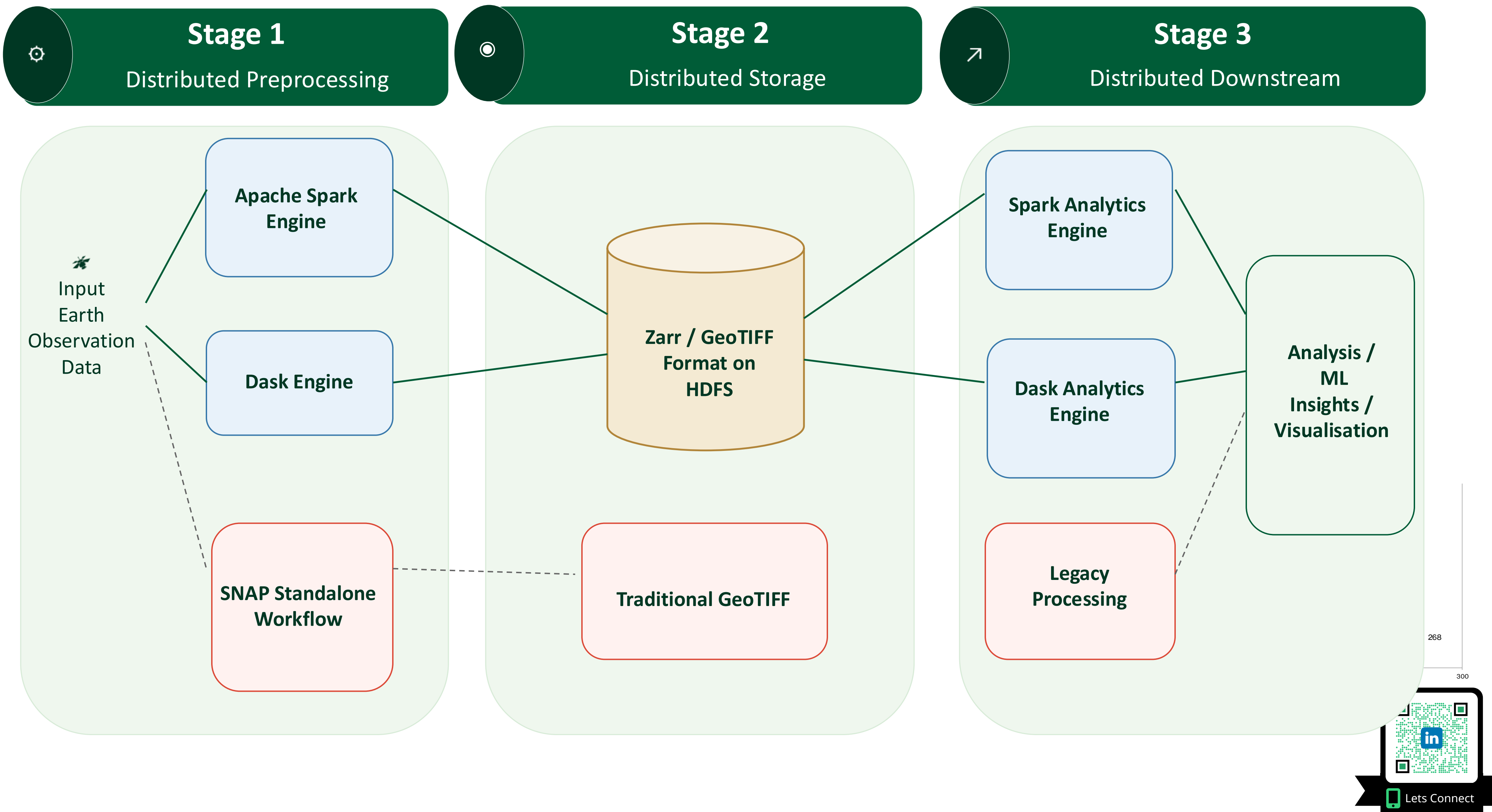


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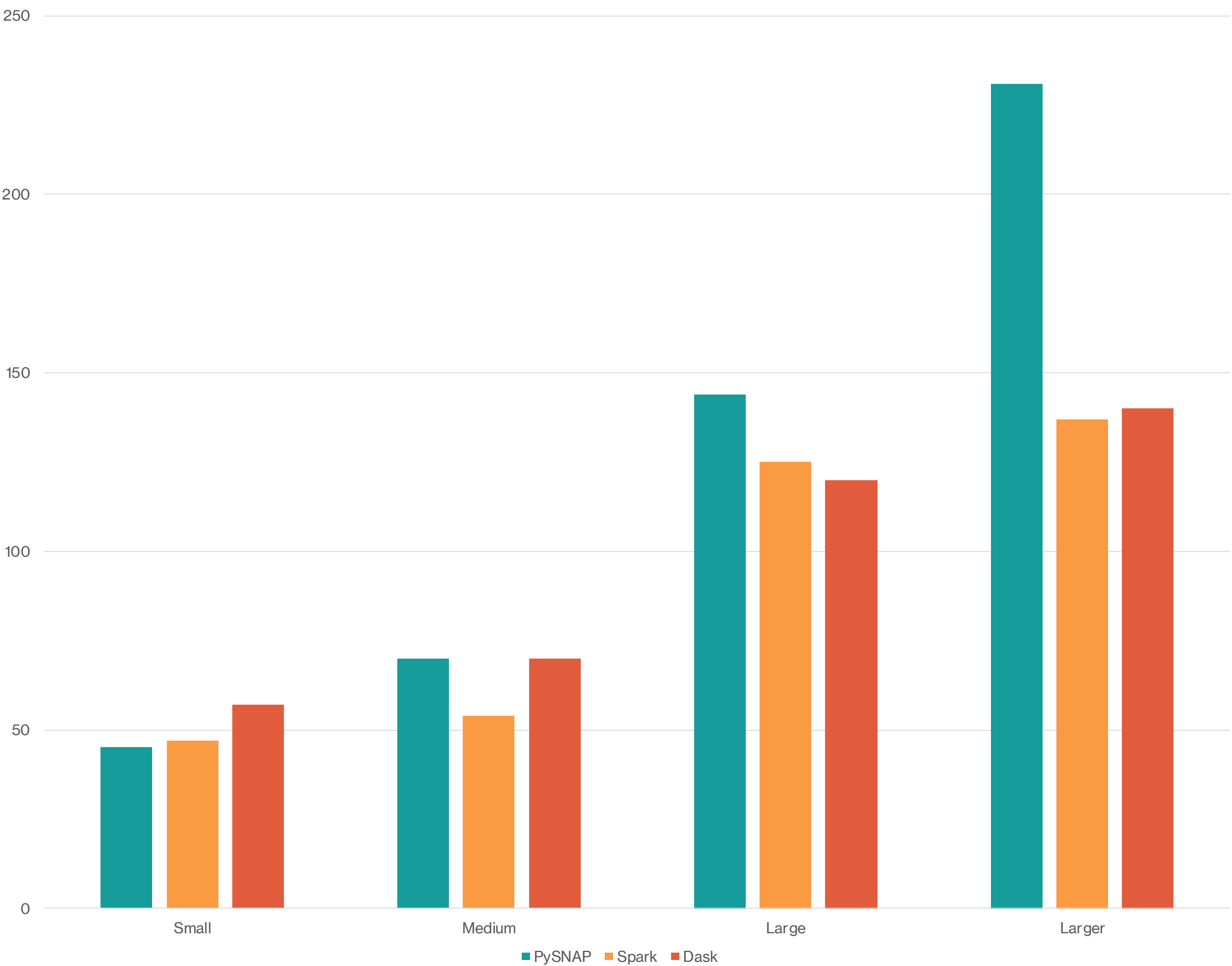
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Preprocessing Time (seconds)

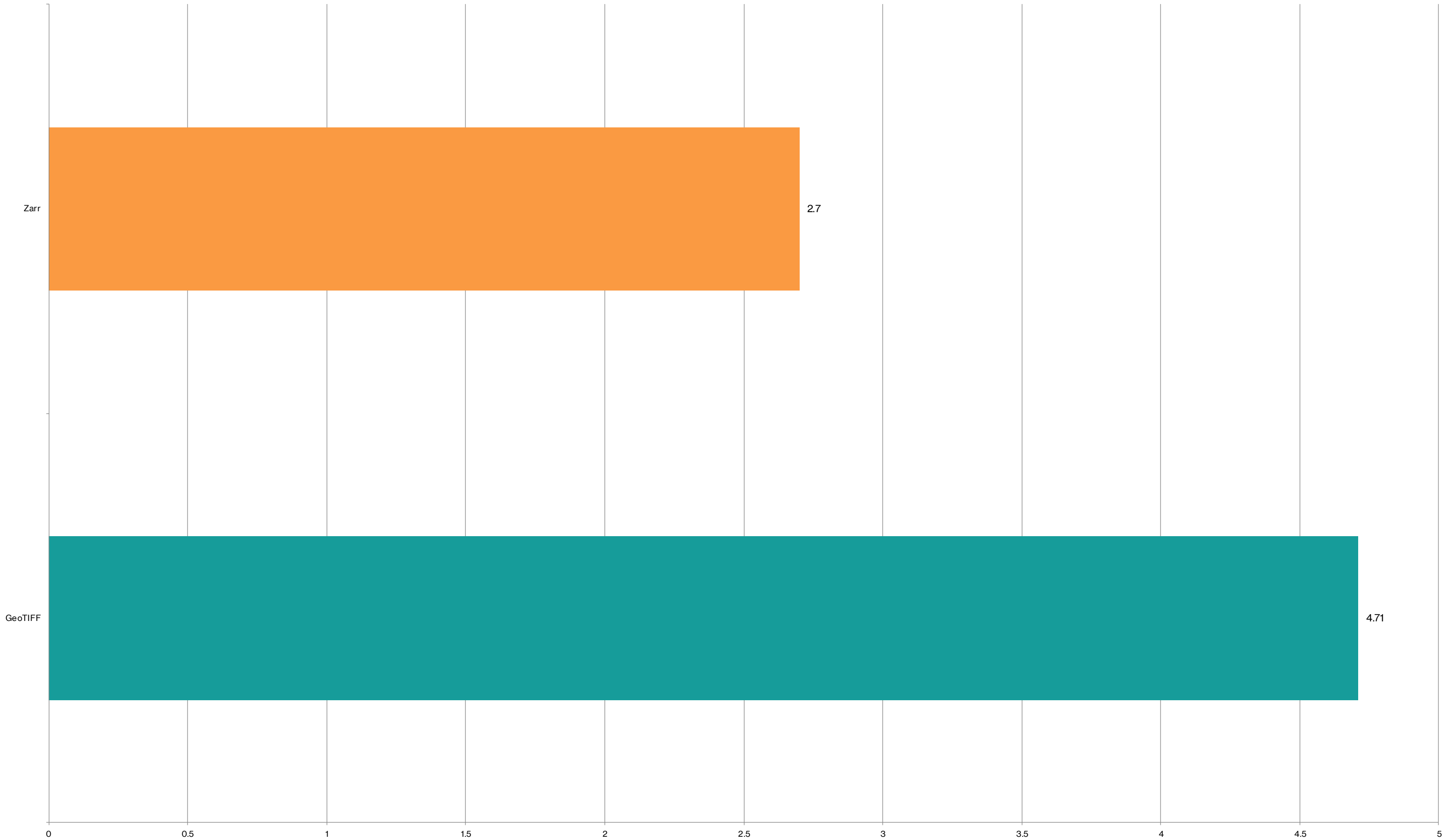


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Storage



42.4%
Space
Saved



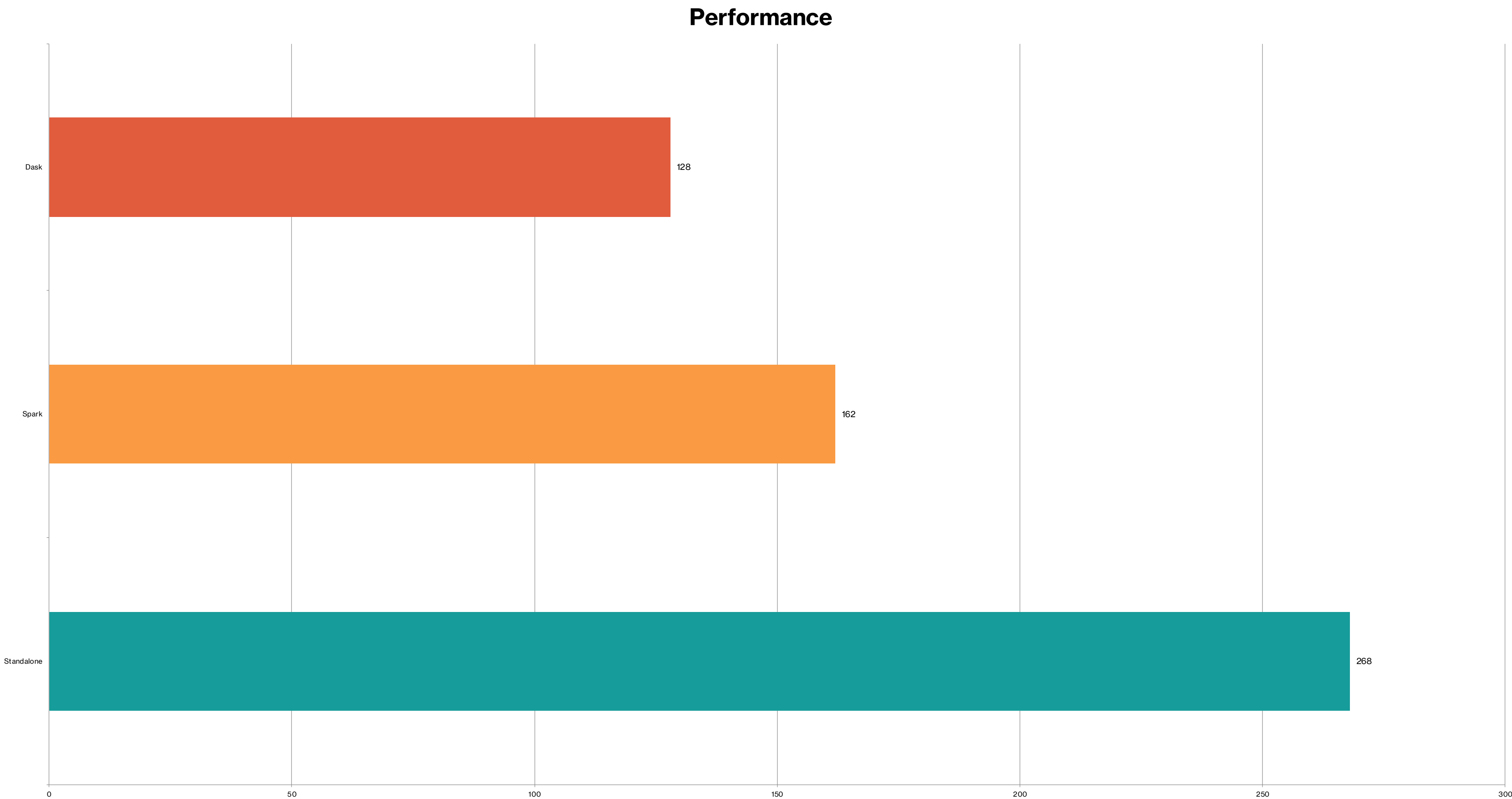
Lets Connect



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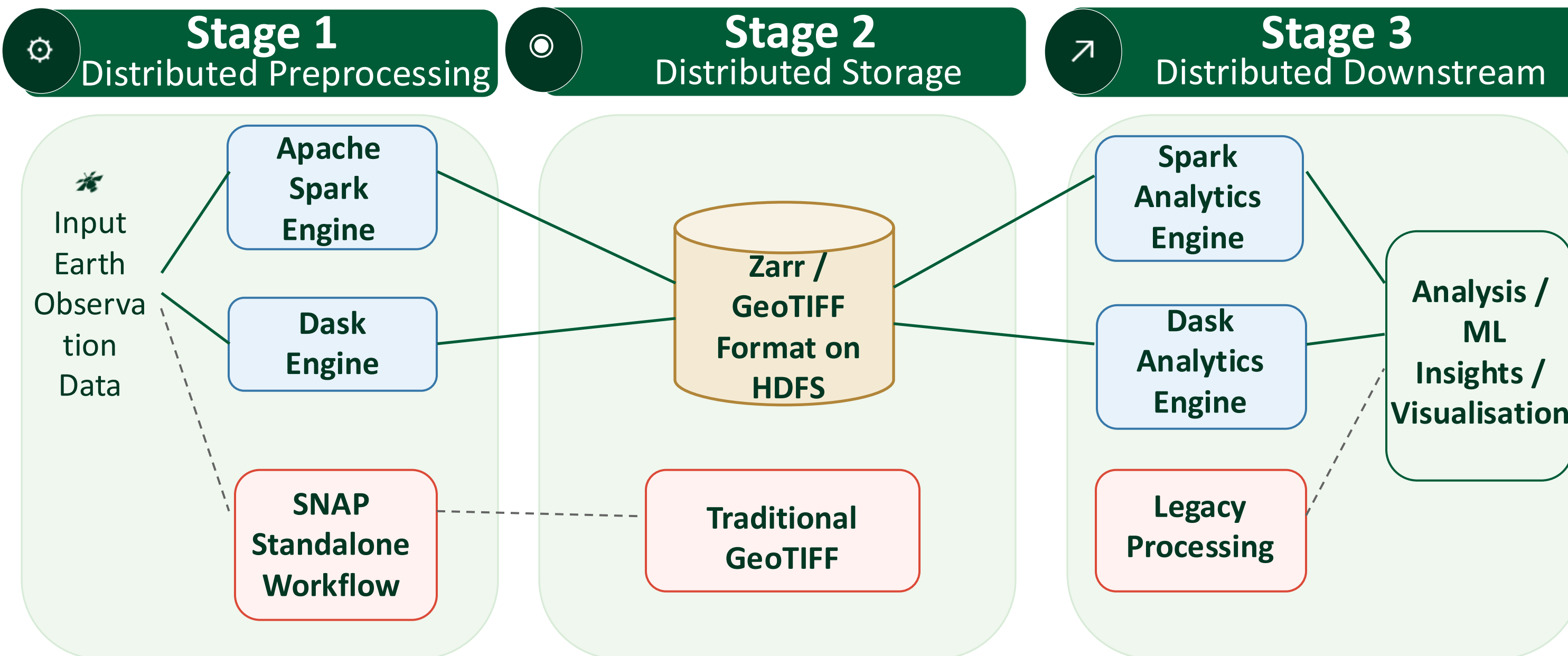
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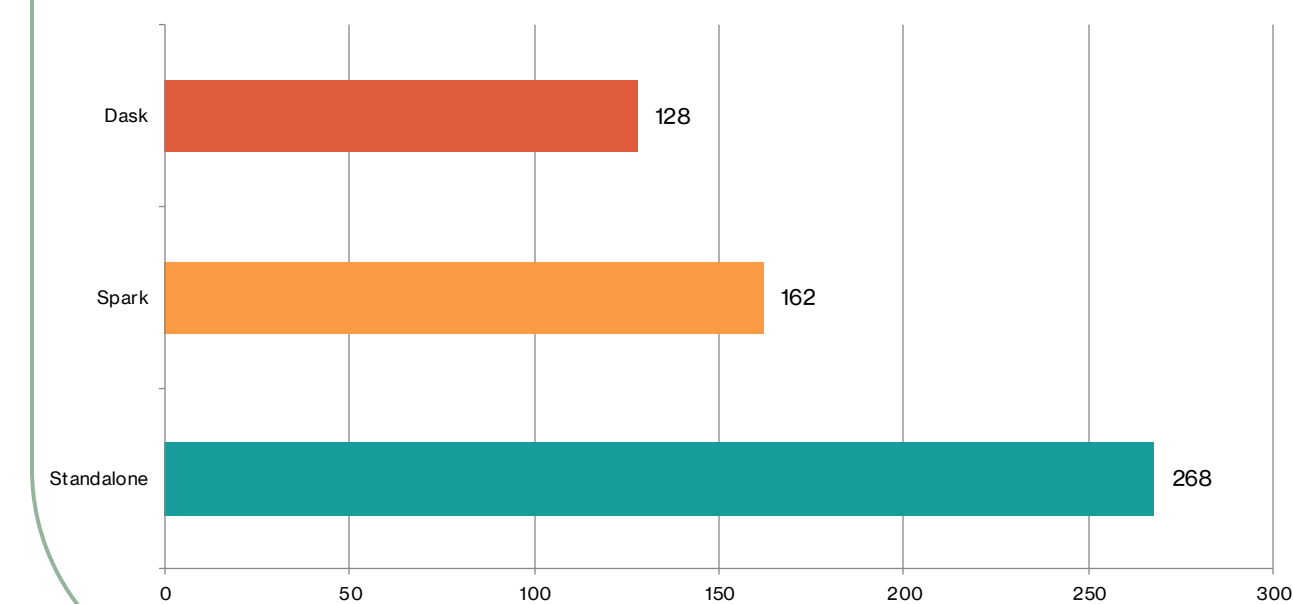
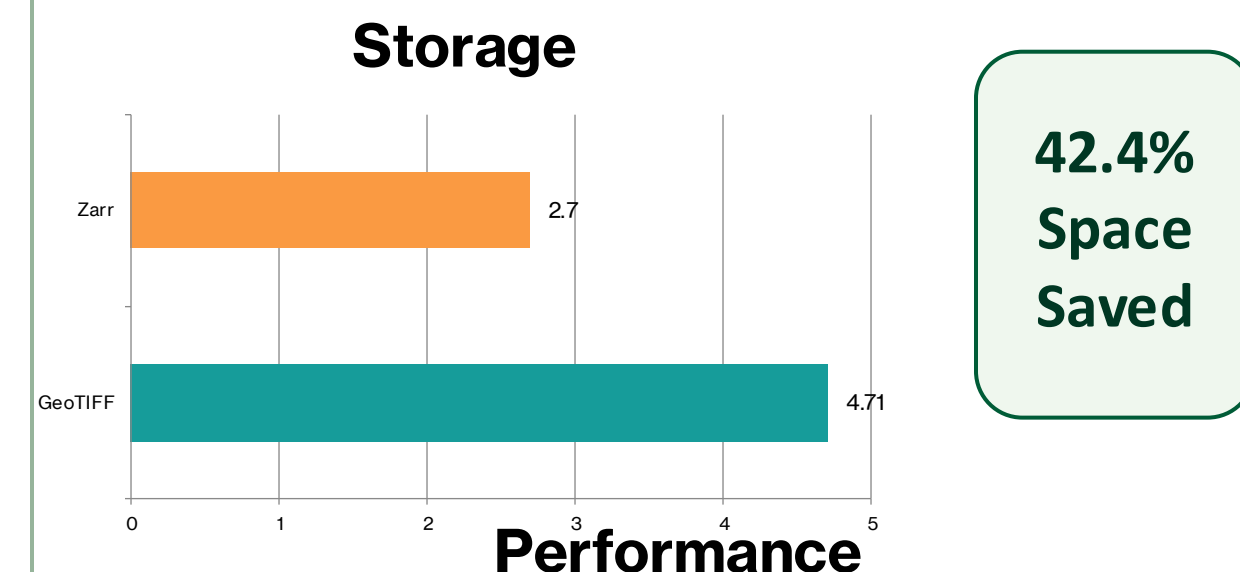
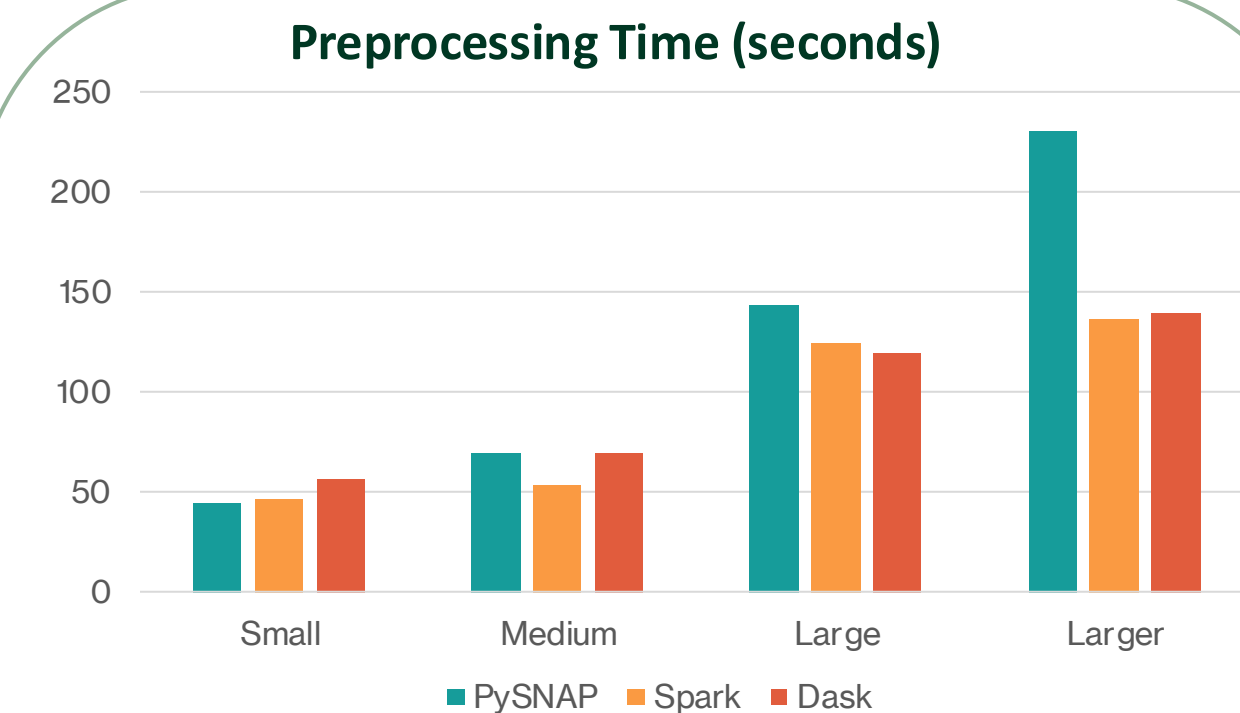
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Supervisor – Dr. Vahid Akbari



Results Overview



Key Results

Feasible Distributed Processing

End-to-end workflow successfully implemented using Spark & Dask

42.4% Reduction in Storage Footprint

Zarr format reduces storage requirements compared with traditional GeoTIFF.

Successful Downstream Analysis

Distributed analytics enables change detection and insights generation.

Conclusion

Feasibility of Big Data to distribute the processing, storage, and analysis of Sentinel-1 Synthetic Aperture Radar data demonstrated, providing a foundation for scalable remote-sensing workflows.

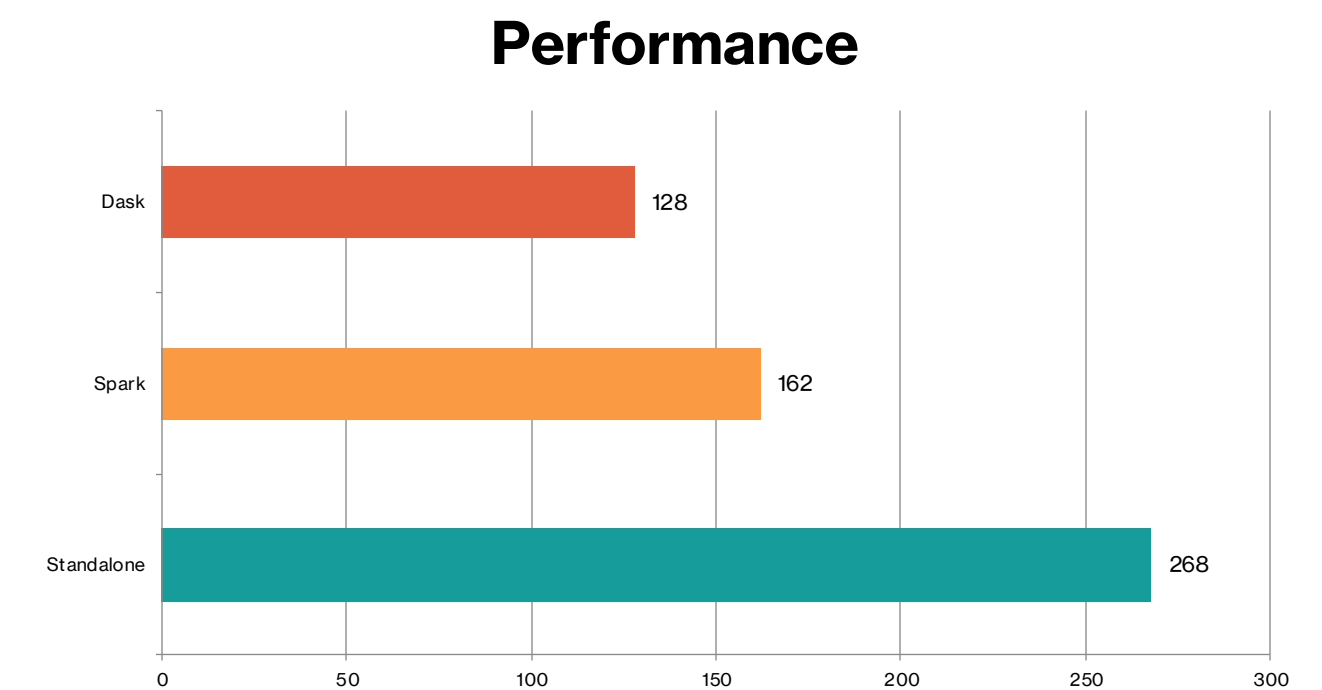


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Thank You



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